Needle in the Real World

How accurate are LLMs in reasoning over unstructured data, especially when confronted with information that may not be in their training data? The existing needle-in-the-haystack benchmarks are a good start, but they are a bit limited for two reasons:

1. The documents are likely to be in the LLMs' training set (e.g., Paul Graham's essays)
2. The questions and inserted facts are not relevant to any particular task

So, I decided to make my own small task and evaluate the models!

The dataset consists of ~85 transcripts of video calls between doctors and their patients, pulled from `train.csv` and `valid.csv` in this repository.

I ask 3 questions--a simple question, medium-difficulty question, and a hard question--to 8 instruction-tuned models:

Model	Input Token Limit	Last Updated
OpenAI GPT-3.5-Turbo-16k	16k	July 2023
OpenAI GPT-4 11-06 preview	128k	November 2023
Anthropic Haiku	200k	March 2024
Anthropic Sonnet	200k	February 2024
Anthropic Opus	200k	February 2024
Gemini Pro	32k	February 2024
Mistral-7B	128k	March 2024
Gemma-7B	8k	March 2024
DBRX-Instruct	32k	March 2024

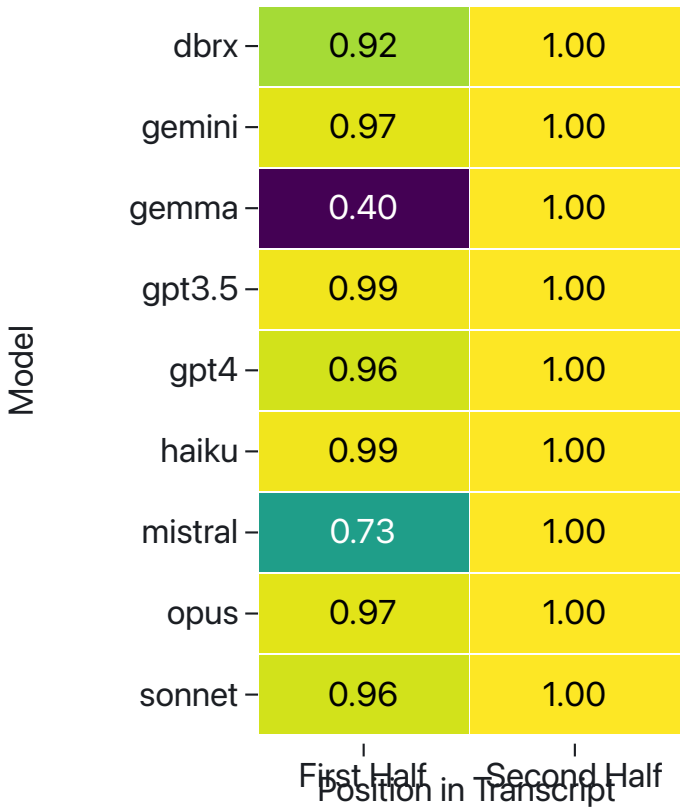
In this report, I show each model's performance, as well as the responses for each question. You can slice and dice the responses for each question, as well as hover over the the scatter plots to see models' answers vs the reference answer. At the end, I run the synthetic needle-in-a-haystack evals.

Note that I manually created the reference answer, with a *lot* of data wrangling, so there are bound to be mistakes. I'm not trying to have perfect data; that would take

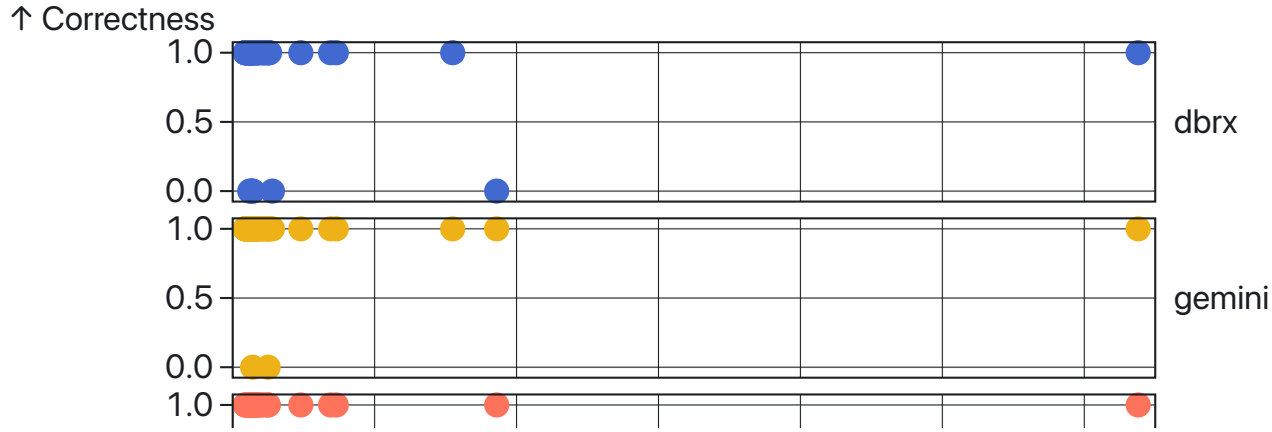
there are bound to be mistakes. I'm not trying to have perfect data, that would take ages :-)

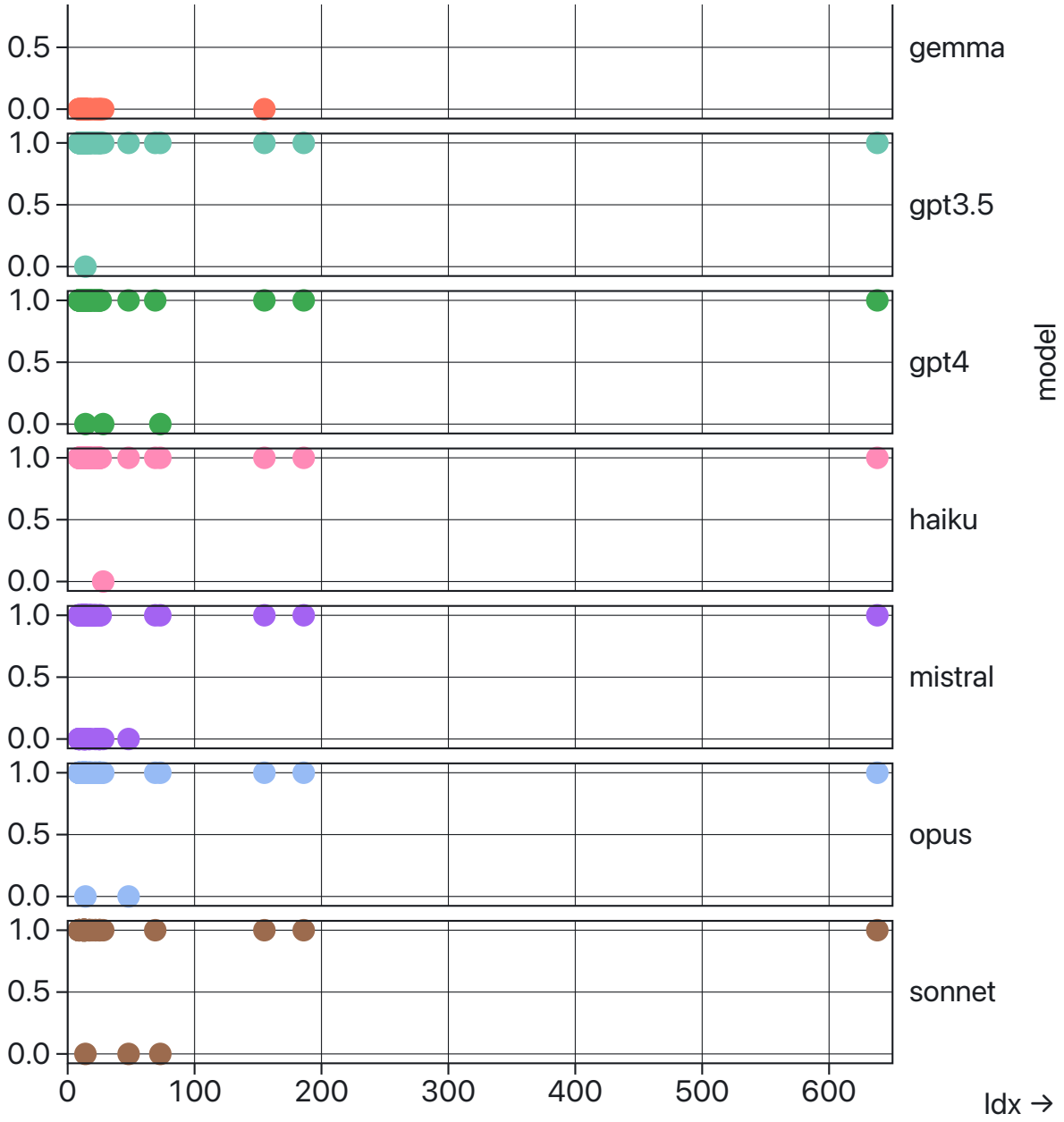
Simple: What is the first name of the patient?

Model Accuracy



Correctness vs. Position of Reference Answer in Transcript





<detached>

<detached>

Takeaways

- gemma is pretty bad; the rest of the models are good
- typically the name is mentioned in the first couple sentences of the transcript. if not, the name is likely mentioned in the end of the transcript (e.g., "Bye [name], have a good day")
- DBRX gives super long answers?
- this is the most basic of questions to make sure the pipeline works, and TLDR: it works

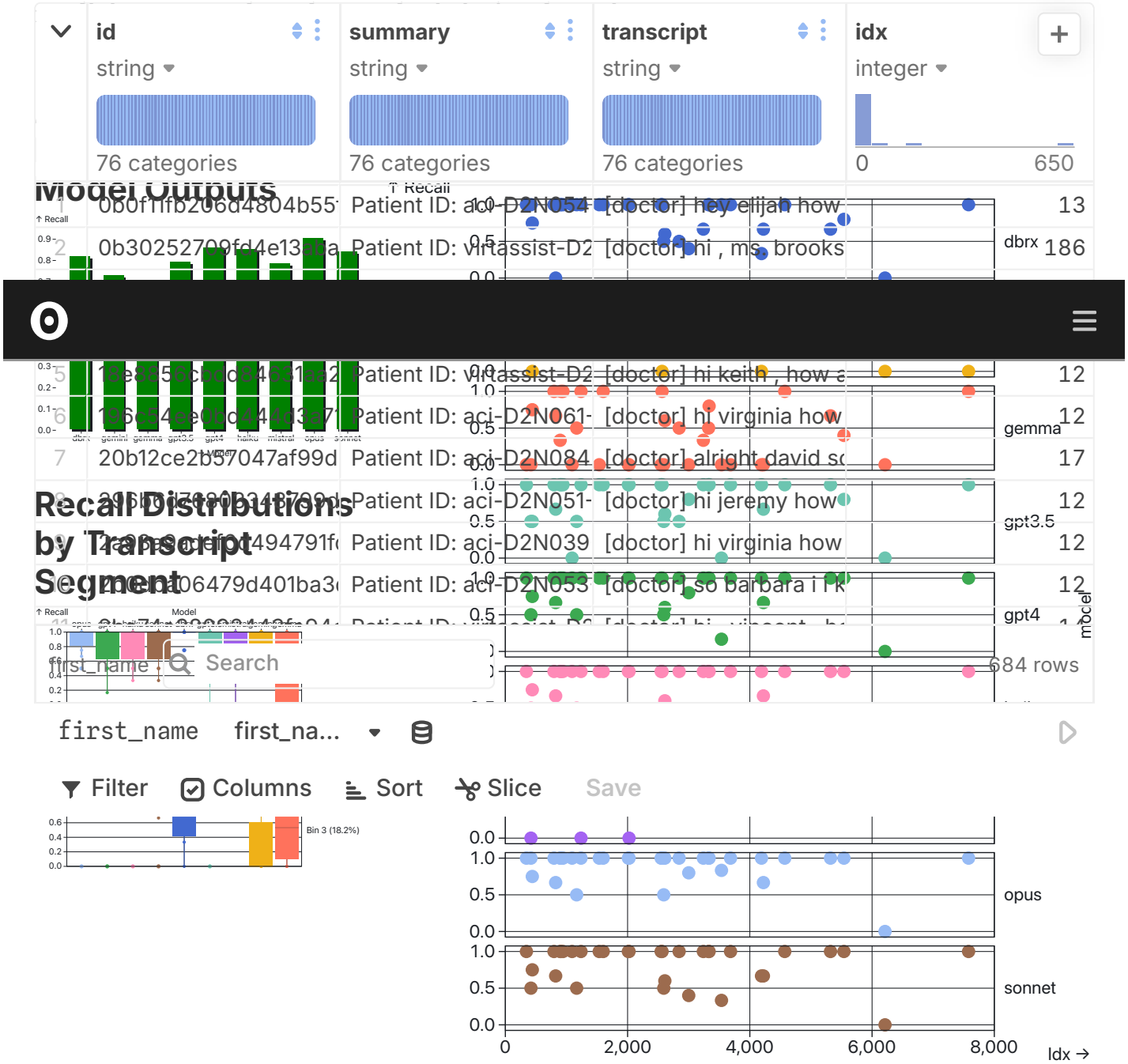
Full Table of Responses

Medium: What medications is the patient currently taking?

taking?

I realize this question is grammatically incorrect; oh well.

Recall is defined as the proportion of words in the reference answer that sound like medicine names and are longer than 5 characters, which are also found in the model's output. I did a lot of manual blacklisting of words/curation to get words that sound like



Takeaways

NOTE: The idx/position here is the average of the positions of each medication in the transcript, so that's why you see floats

- The task, listing medications, simply asks the model to regurgitate various medications from the transcript
- Medications currently taken are distributed mostly in the first third of the transcript
- Opus is slightly better than GPT-4 in my metrics. But if you hover over the bad answers in the scatterplot for GPT-4, you'll find that GPT-4 rephrases the medications slightly
- Claude models follow a super consistent and slightly more verbose structure?
- Personally I would prefer Opus to GPT-4 here
- All models degrade in performance when the answer is in the second and third segments of the transcript (note that segments are equally sized)
- Gemini fails to return any meaningful result (e.g., it returns empty string or N/A) for several of the calls
- Haiku and Sonnet look eerily similar in performance (but clearly have different responses)
- DBRX has super long responses (and parts of the first part of the response seem to be half system prompt? maybe i prompted incorrectly)

Full Table of Responses

<detached>

<detached>

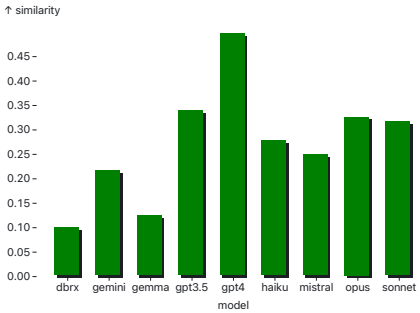
<detached>

<detached>

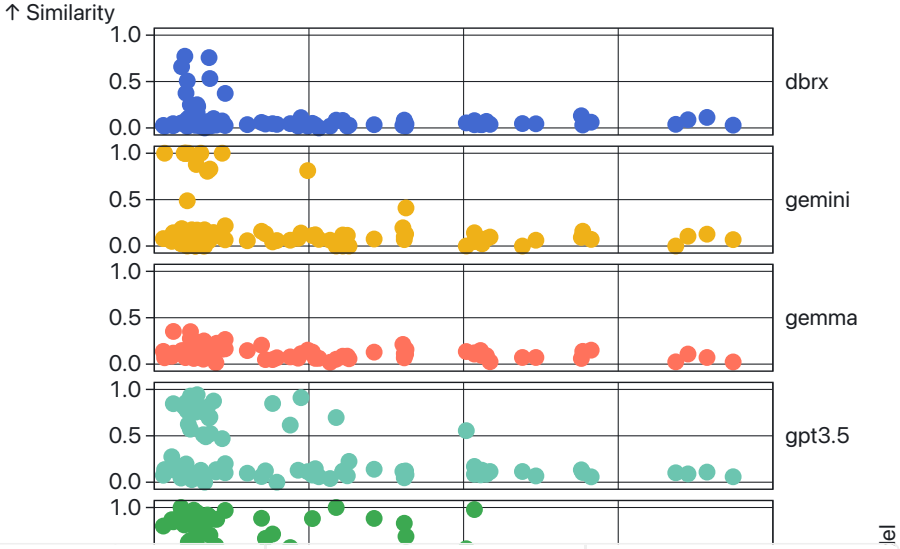
Hard: What is the longest sentence the patient said?

Similarity is Jaccard similarity

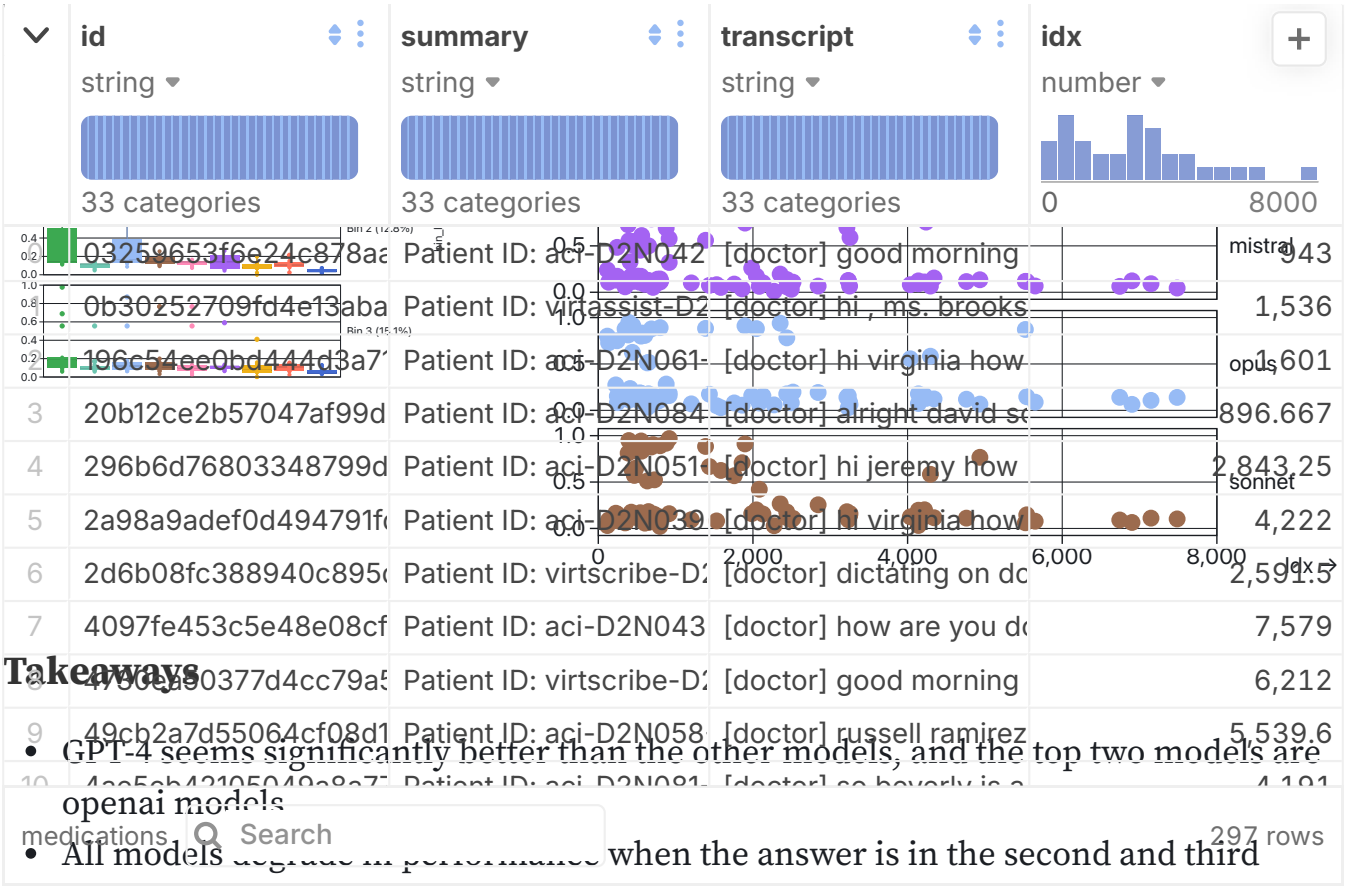
Avg Similarity Between Outputs and Reference Answer



Similarity vs. Position of Reference Answer in Transcript



Similarity Distributions by



	id	summary	transcript	idx
	string	string	string	number
	33 categories	33 categories	33 categories	0 – 8000
1	03258653f6e24c878aa	Patient ID: aci-D2N042	[doctor] good morning	943
2	0b30252709fd4e13aba	Patient ID: virtassist-D2	[doctor] hi , ms. brooks	1,536
	196c54ee0bd444d3a7	Patient ID: aci-D2N061-	[doctor] hi virginia how	1,601
3	20b12ce2b57047af99d	Patient ID: aci-D2N084	[doctor] alright david s	896.667
4	296b6d76803348799d	Patient ID: aci-D2N051	[doctor] hi jeremy how	2,843.25
5	2a98a9adef0d494791f	Patient ID: aci-D2N039	[doctor] hi virginia how	4,222
6	2d6b08fc388940c895	Patient ID: virtscribe-D2	[doctor] dictating on dc	2,591.5
7	4097fe453c5e48e08cf	Patient ID: aci-D2N043	[doctor] how are you d	7,579
8	4f9ca90377d4cc79a5	Patient ID: virtscribe-D2	[doctor] good morning	6,212
9	49cb2a7d55064cf08d1	Patient ID: aci-D2N058	[doctor] russell ramirez	5,539.6
10	4ac5cb42105049a8a7	Patient ID: aci-D2N081-	[doctor] so beverly is a	4,191

medications Search 297 rows

medications medicati... Filter Columns 10 Sort Slice Save

Takeaways

- GPT-4 seems significantly better than the other models, and the top two models are openai models
- All models degrade in performance when the answer is in the second and third

flip the patient-doctor labels and rerun this experiment :-)

- Mistral seems to attach a number of words to its response? no idea why, but these are not always correct
- The better-performing models give a consistent structure
- Gemini is just...bad?
- DBRX is *really* bad...maybe i prompted it wrong? i did copy the system prompt they gave in huggingface

Full Table of Responses

<detached>

<detached>

A Final Test: Synthetic Eval

How do the models perform on the standard needle-in-the-haystack eval, where the haystack is our transcripts?

To test this, we leverage a similar experimental setup to what's been done before, but we focus on retrieving a single needle.

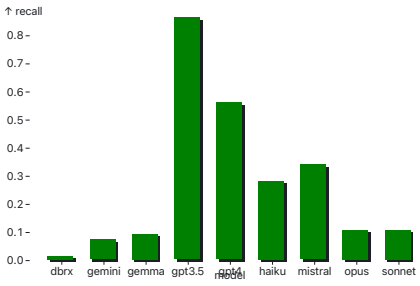
For each transcript, we randomly select 3 toppings from a list of 10 pizza toppings, and concatenate them into a sentence. E.g., *The secret ingredients needed to build the perfect pizza are: Espresso-soaked dates, Lemon and Goat cheese.*

Next, we (uniformly) randomly pick an index in the document to insert this sentence.

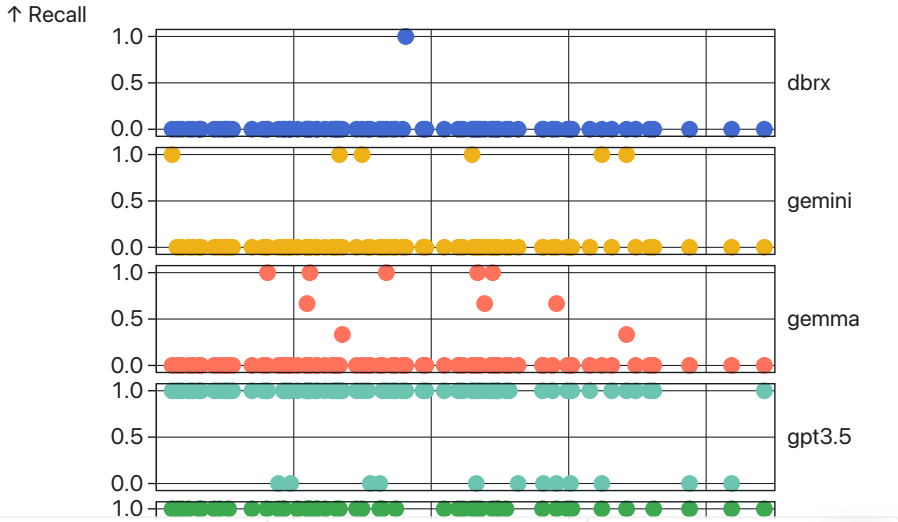
Then, we ask each model the following question: *What are the secret ingredients needed to build the perfect pizza?*

For each model and document, we compute the recall of secret ingredients (i.e., 0, 0.33, 0.67, or 1.0).

Avg Recall of Needles



Recall vs. Position of Needle in Transcript



Recall Distributions by Transcript Segment

	id	summary	transcript	idx
	string	string	string	integer
	86 categories	86 categories	86 categories	0 — 7500
0	03259653f6e24c878aa	Patient ID: aci-D2N042	[doctor] good morning	642
1	0341ef9ec1f14d77b1d59	Patient ID: aci-D2N056	[patient] alright thanks	4,140
2	0b0f11fb206d4804b55	Patient ID: aci-D2N054	[doctor] hey elijah how	878
3	0b30252709fd4e13aba	Patient ID: virtassist-D2	[doctor] hi , ms. brooks	440
4	10cc26d3967a4af0bd4	Patient ID: virtscribe-D	[doctor] patient , bruce	4,935
5	1488167ac0834b9085a	Patient ID: aci-D2N057	[patient] hi good aftern	5,650
6	18e8856cbdd84631aa2	Patient ID: virtassist-D2	[doctor] hi keith , how's	1,860
7	196c54ee0bd444d3a7	Patient ID: aci-D2N061-	[doctor] hi virginia how	243
8	1e7e85d72f464e0e968	Patient ID: virtassist-D2	[doctor] hi ms. hernand	466
9	20b12ce2b57047af99d	Patient ID: aci-D2N084	[doctor] alright david s	246
10	296b6d76803348799d	Patient ID: aci-D2N051-	[doctor] hi jeremy how	1,387

sentences Search 774 rows

sentences longest_...

Filter Columns Sort Slice Save

Takeaways

NOTE: The max idx is larger here (~13k) because all transcripts contain an answer. In previous questions, the super-long transcripts didn't contain any answers, so they weren't included in the analysis.

- None of the models are perfect! But other peoples' evals show perfect results? I guess this medical transcript data wasn't in the training set, so who knows what the result would be.
- GPT 3.5 is the best!! That is quite surprising. I double and triple checked my code; maybe there is a bug. But I couldn't find one. Also the GPT-4 responses are quite verbose, which aligns with my expectations.
- The models either get all 3 or get none, with the exception of Gemma, which occasionally retrieves 2 of the 3??
- Claude models are worse than Mistral
- Surprisingly Mistral gets the ingredients, but you can see that its responses are a bit verbose
- Gemini is quite bad, but many of the outputs are empty so that explains the poor performance
- Sometimes the models are self-aware, and say that the transcript is not about pizza :-)
- DBRX completely flops on all but 1!! It does not even output anything...again, maybe i'm sampling wrong

Full Table of Responses

<detached>

<detached>

<detached>

Needle in the Synthetic World

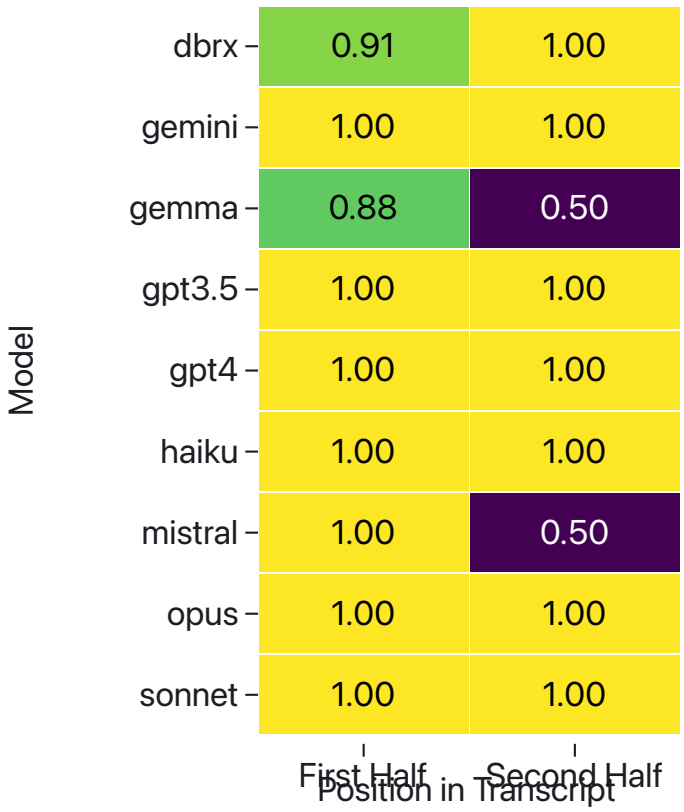
What happens when we run the same experiments as above, but we use GPT-4 to generate *synthetic* transcripts between patients and providers? I generated 35 synthetic transcripts with a prompt that included a random patient name (from some list of names I queried GPT-4) and random set of medications (again, from a list of medications I queried GPT-4). The prompt to generate synthetic transcripts is as follows:

```
Here is an example of a transcript of a patient's visit to the doctor:\n\n{
synthetic transcript containing at least {num_words} words, representing a
{patient_name}, and they are taking the medications {medications}. Make sur
medications (and dosages) they are taking. Return ONLY the synthetic transc
```

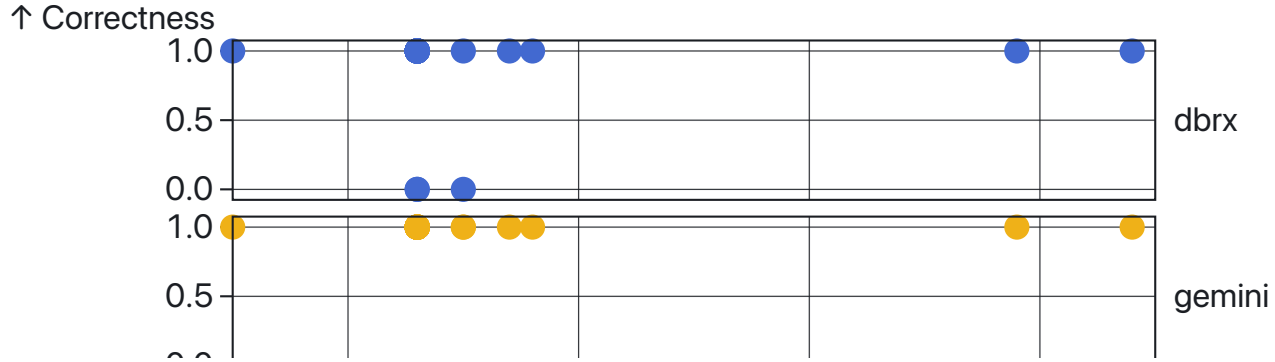
For each of the 35 LLM calls to generate a synthetic transcript, I randomly selected a real transcript to provide as the example transcript (and `num_words`) in the prompt.

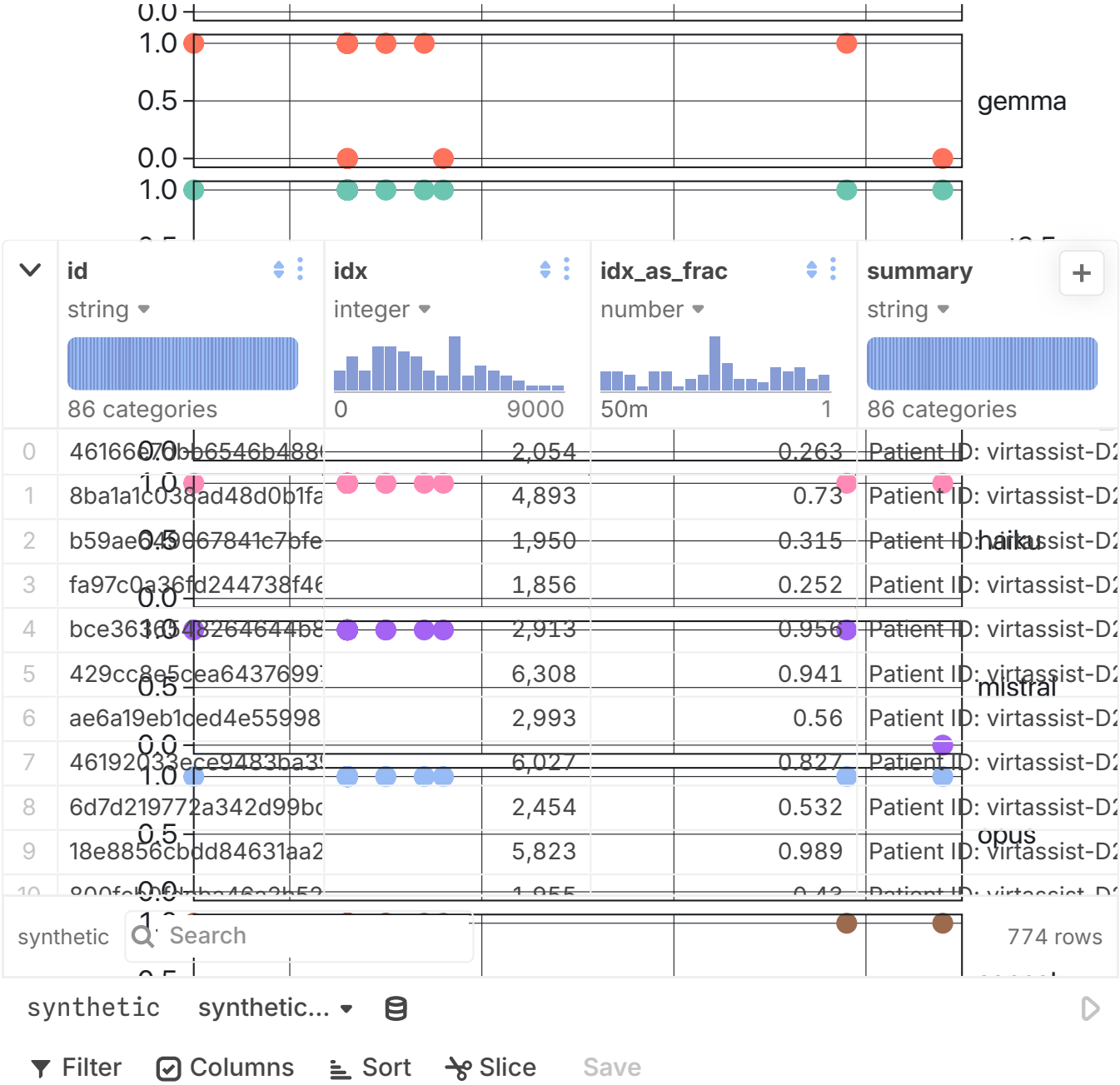
What is the first name of the patient?

Model Accuracy on Synthetic Transcripts (GPT-4 generated)



Correctness vs. Position of Reference Answer in Transcript (for Synthetically-Generated Transcripts)





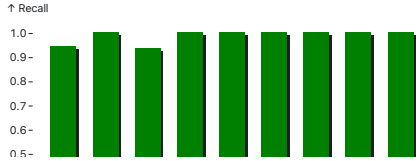
Takeaways

This is near 100% accuracy for all models.

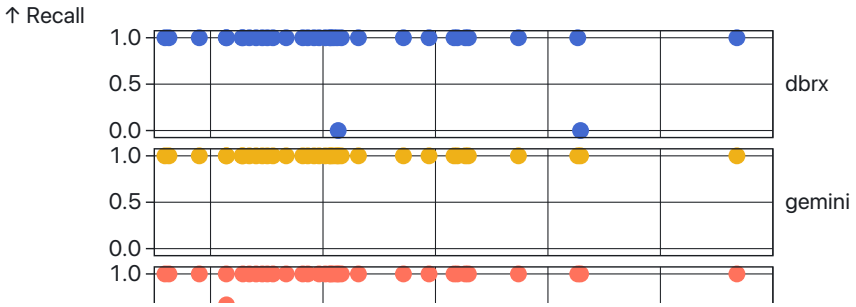
<detached>

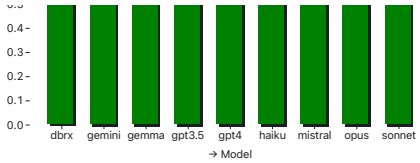
Medium: What medications is the patient currently taking?

Avg Recall of Reference Answer in Model Outputs (on Synthetic Transcripts)

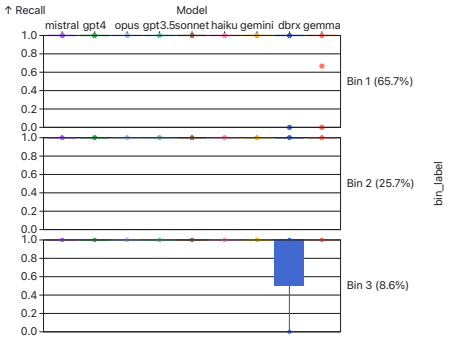


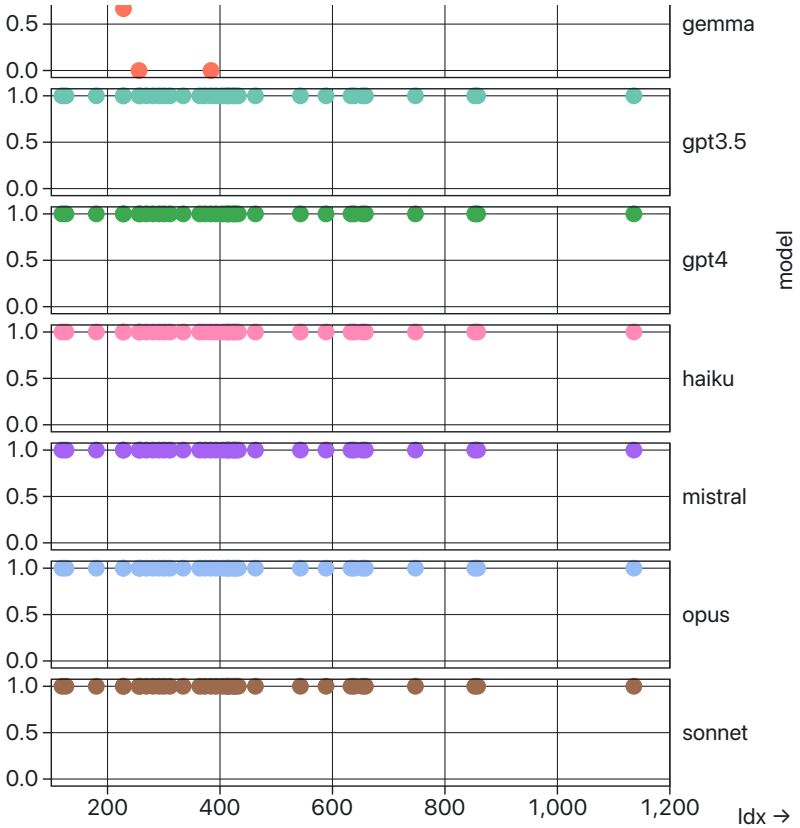
Recall vs. Position of Reference Answer in Transcript (on Synthetic Transcripts)





Recall Distributions by Transcript Segment (on Synthetic Transcripts)





Takeaways

This is also basically 100% accuracy for all models.

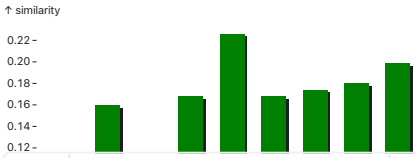
<detached>

<detached>

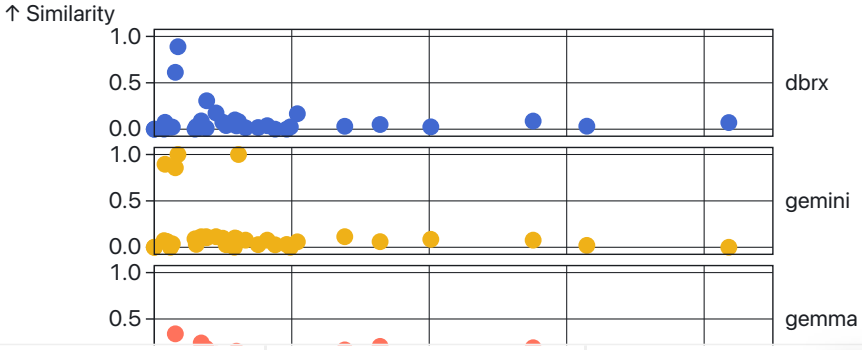
<detached>

Hard: What is the longest sentence the patient said?

Avg Similarity Between Outputs and Reference Answer (on Synthetic Transcripts)

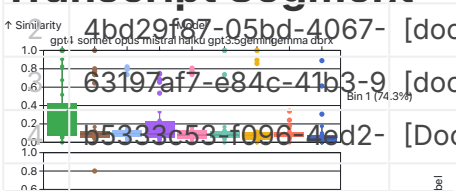


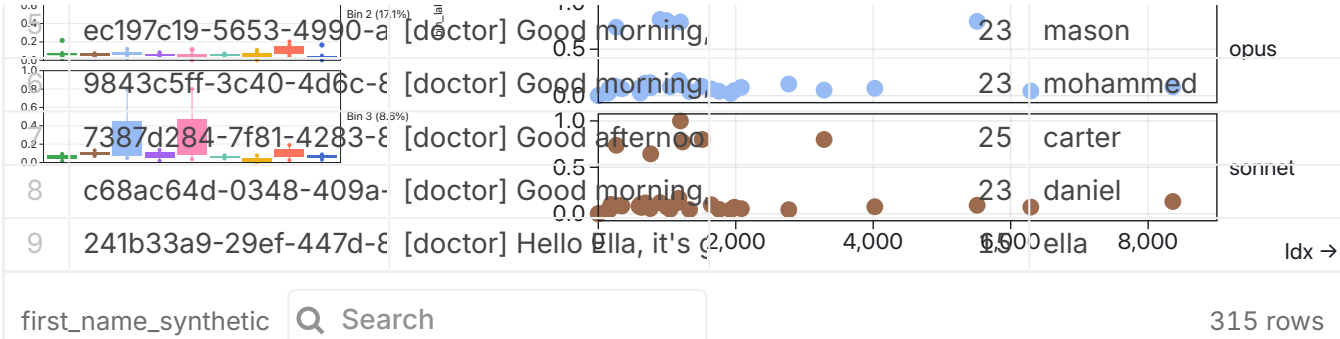
Similarity vs. Position of Reference Answer in Transcript (on Synthetic Transcripts)



	id	transcript	idx	first_name
	string	string	integer	string
	35 categories	35 categories	14 — 56	24 categories
0	27eef273-708e-4371-a	[doctor] Good morning,	23	mohammed
1	d6880b94-0c19-440d-	[doctor] Good morning,	23	charlotte
2	4bd29f87-05bd-4067-	[doctor] Good morning,	23	aiden
3	63197af7-e84c-41b3-9	[doctor] Good morning,	23	sophia
4	b5333c53-f096-4ed2-	[Doctor] Good morning	23	liam

Similarity Distributions by Transcript Segment

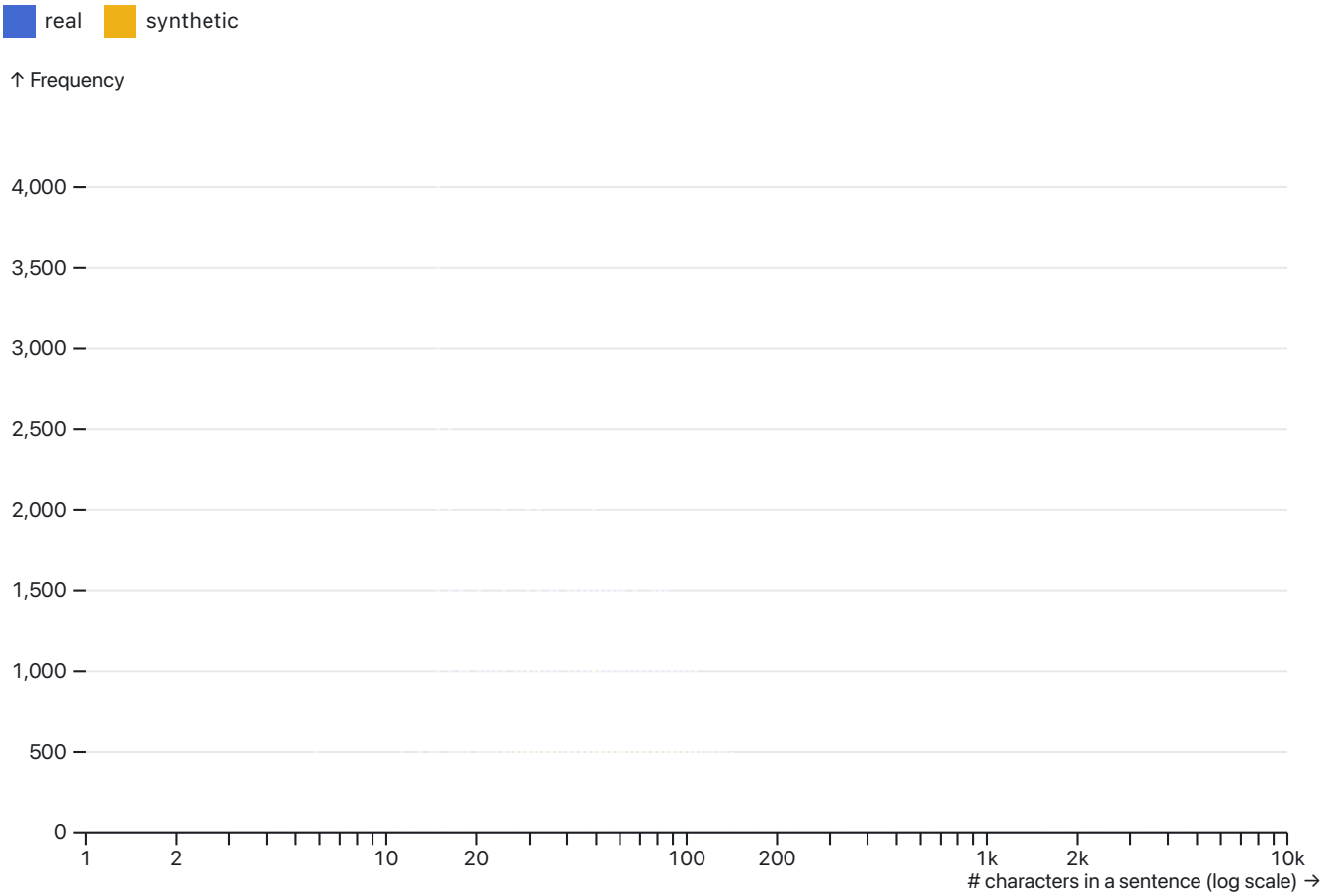




Takeaways

more or less of the same length, so it's actually pretty difficult to get exactly the right longest sentence (see the histogram below, where there are lots of blue outliers). Also, I guess I wasn't precise in the question--is it the longest sentence by # words or by # characters? My "ground truth" is computed with # characters.

Histogram of Sentence Lengths in Transcripts



<detached>

<detached>

<detached>

	id	transcript	idx	medications
	string	string	number	string
	35 categories	35 categories	100 1150	35 categories
0	27eef273-708e-4371-a	[doctor] Good morning,	425	Adalimumab
1	d6880b94-0c19-440d-	[doctor] Good morning,	658.5	Dabigatran Aspirin
2	4bd29f87-05bd-4067-	[doctor] Good morning,	1,136	Hydrochlorothiazide Li
3	63197af7-e84c-41b3-9	[doctor] Good morning,	364	Amlodipine Lisinopril
4	b5333c53-f096-4ed2-	[Doctor] Good morning	633	Azithromycin
5	ec197c19-5653-4990-a	[doctor] Good morning,	419	Paracetamol Codeine
6	9843c5ff-3c40-4d6c-8	[doctor] Good morning,	653	Cetirizine
7	7387d284-7f81-4283-8	[doctor] Good afternoo	432.5	Pregabalin Duloxetine
8	c68ac64d-0348-409a-	[doctor] Good morning,	257.5	Infliximab Methotrexate
9	24fb83a9-29ef-447d-c	[doctor] Hello Lila, it's	228.333	Digoxin Furosemide Sp
10	[illegible]	[illegible]	384	Ibuprofen

medications_synthetic Search 315 rows

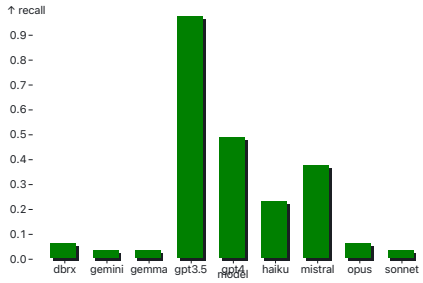
medications_synthetic medicati...

Filter Columns 9 Sort Slice Save

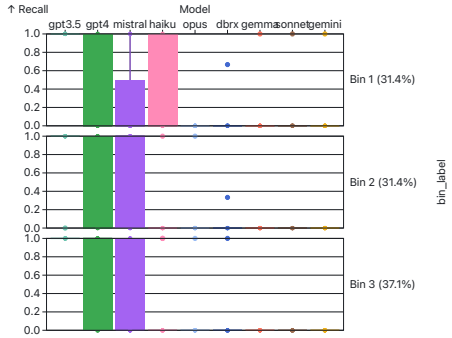
Synthetic Eval on Synthetic Transcripts :-)

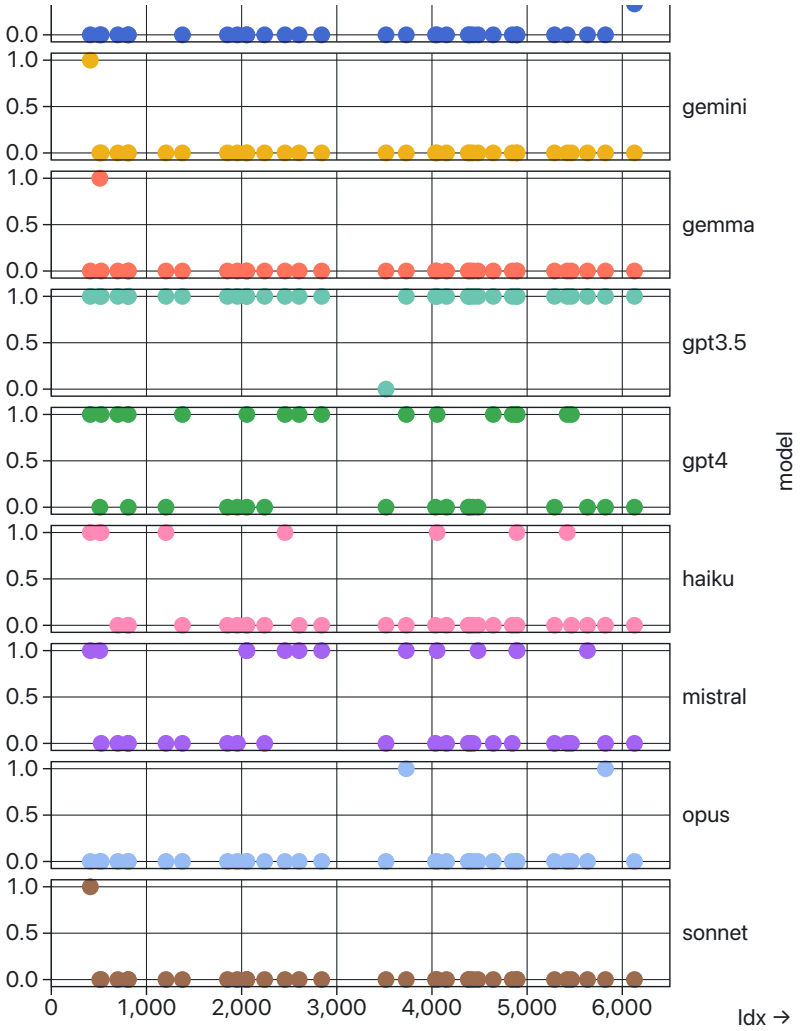
We use the same needle in a haystack pizza related eval.

Transcripts)



Recall Distributions by Transcript Segment (on Synthetic Transcripts)





Takeaways

Ok, this is really weird. Performance got worse for all models except GPT-3.5 and Mistral. Can't explain why...

Inspecting the outputs, it seems like most outputs where where recall = 0 reference something like "The conversation is focused on a conversation between a doctor and patient..."

I wonder if, because these synthetic transcripts are, by definition, in-distribution for LLMs, inserting some phrase related to the secret ingredients for pizza is quite "jarring" for the models." But who knows.

<detached>

<detached>

<detached>

	id	transcript	idx	longest_sente...
	string	string	integer	string
	35 categories	35 categories	0 8500	34 categories
0	27eef273-708e-4371-a	[doctor] Good morning,	2,081	Sure, it's mainly here ir
1	d6880b94-0c19-440d-	[doctor] Good morning,	3,285	For breakfast, I usually

2	4bd29f87-05bd-4067-	[doctor] Good morning,	262	I've been having some
3	63197af7-e84c-41b3-9	[doctor] Good morning,	5,512	Just to recap, I'll keep t
4	b5333c53-f096-4ed2-	[Doctor] Good morning	0	<empty string>
5	ec197c19-5653-4990-a	[doctor] Good morning,	755	I've felt a bit drowsy sc
6	9843c5ff-3c40-4d6c-8	[doctor] Good morning,	1,979	Not really, I've been ke
7	7387d284-7f81-4283-8	[doctor] Good afternoo	1,195	I've been experiencing
8	c68ac64d-0348-409a-	[doctor] Good morning,	1,045	The pain is less intense
9	241b33a9-29ef-447d-8	[doctor] Hello Ella, it's g	235	I've been taking the me

sentences_synthetic Search 315 rows

sentences_synthetic longest_...

Filter Columns Sort Slice Save

	id string 35 categories	idx integer 0 – 6500	idx_as_frac number 50m – 1	summary other all values: "other"
0	27eef273-708e-4371-a	2,054	0.384	<empty string>
1	d6880b94-0c19-440d-	4,893	0.881	<empty string>
2	4bd29f87-05bd-4067-	4,644	0.832	<empty string>
3	63197af7-e84c-41b3-9	3,730	0.587	<empty string>
4	b5333c53-f096-4ed2-	2,841	0.543	<empty string>
5	ec197c19-5653-4990-a	808	0.13	<empty string>
6	9843c5ff-3c40-4d6c-8	410	0.081	<empty string>
7	7387d284-7f81-4283-8	699	0.123	<empty string>
8	c68ac64d-0348-409a-	5,823	0.88	<empty string>
9	241b33a9-29ef-447d-8	1,955	0.346	<empty string>
10	0eef401b-7061-4918-8	6,128	0.595	<empty string>

synthetic_synthetic Search 315 rows

synthetic_synthetic synthetic...

Filter Columns Sort Slice Save

Purpose-built for displays of data

Observable is your go-to platform for exploring data and creating expressive data visualizations. Use reactive JavaScript notebooks for prototyping and a collaborative canvas for visual data exploration and dashboard creation.

Try it for free →

Learn more